REAL NUMBERS

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Lecture Outline

1. Numbers
2. Standard subsets
   - Natural Numbers
   - Integers
   - Rational Numbers
   - Irrational and Real Numbers
3. Some Properties of Numbers
4. Relations on $\mathbb{R}$
   - Inequalities
5. Absolute value



Outline of Presentation

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Introduction

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Different quantities may require different kinds of numbers to describe them leading to classification of numbers.



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Natural Numbers

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$\mathbb{N}$ admits some binary operations including **addition** and **multiplication**.



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Theorem

*The set $\mathbb{N}$ is **well-ordered**, that is every subset of $\mathbb{N}$ has a least element.*



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4. Multiplicative identity: There is some $e \in \mathbb{N}$ such that $e \times n = n = n \times e$ for all $n \in \mathbb{N}$.
5. Cancellation: $m = k$ if $m + n = k + n$ or $m \times n = k \times n$.



Integers

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The set of integers denoted by $\mathbb{Z}$ is the set $\{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$. Thus $\mathbb{Z} = \mathbb{Z}^- \cup \{0\} \cup \mathbb{Z}^+$

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$\mathbb{Z}$ admits subtraction operation by the definition $a - b = a + (-b)$. It is evident that $\mathbb{Z}$ is not commutative nor associative under this operation.

Rational numbers

Definition

The set of rational numbers denoted by $\mathbb{Q}$ is the set of all numbers which can be written in the form $\frac{x}{y}$ where $x, y \in \mathbb{Z}$ and $y \neq 0$.

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1. All the properties observed under $\mathbb{Z}$ holds here.
2. Every nonzero rational number has a multiplicative inverse.

Irrational and Real Numbers

Irrational numbers

1. These are decimals which do not fit the description of rational numbers.
2. That is non-repeating nor terminating decimals.
3. Alternatively these are numbers which can not be written as ratio of integers.
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Real Numbers

The set of real numbers denoted $\mathbb{R}$ is the union of $\mathbb{Q}$ and $\overline{\mathbb{Q}}$.



Summary Page

Some

Some of the

standard subsets of $\mathbb{R}$ are:

1. $\mathbb{N} = \{1, 2, 3, \cdots\}$ is the set of natural numbers.
2. $\mathbb{W} = \{0, 1, 2, 3, \cdots\}$ is the set of whole numbers.
3. $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \cdots\}$ is the set of integers.
4. $Q = \{\frac{n}{m} : n, m \in Z, m \neq 0\}$ is the set of rational numbers.
5. $\bar{\mathbb{Q}}$ is the set of irrational numbers or decimals which are not in $\mathbb{Q}$.
6. $\mathbb{R} = \mathbb{Q} \cup \bar{\mathbb{Q}}$ is the set of real numbers.



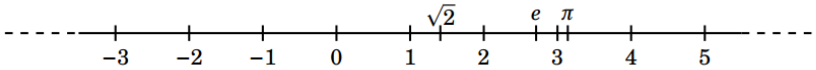
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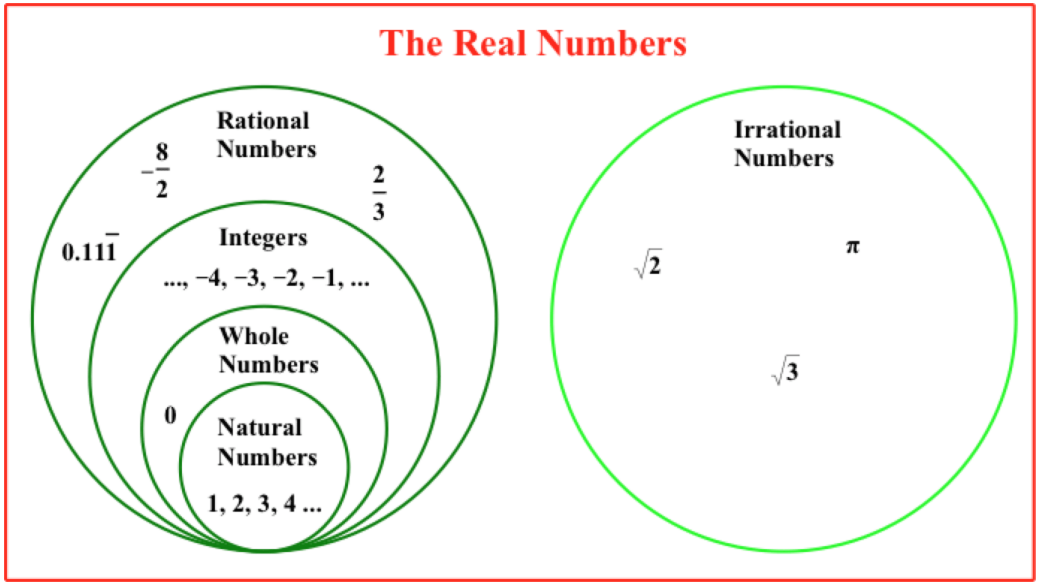
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The Real Numbers
Rational Numbers
$-\frac{8}{2}$
$\frac{2}{3}$
$0.11\overline{1}$
Integers
..., −4, −3, −2, −1, ...
Whole Numbers
0
Natural Numbers
1, 2, 3, 4 ...
Irrational Numbers
$\sqrt{2}$
π
$\sqrt{3}$

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Closure

Definition

Let G be a set and let H be a subset of G. Then, for any two elements $g, h \in H$, then their binary operation $g * h$ will be an element of G, which may or may not be in H. If $g * h \in H$ for all $g, h \in H$, we say that H is closed under the binary operation of G.



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1. The set of natural numbers is closed under $+$ and $\times$, but not closed under $-$ and $\div$.
2. The set of integers is closed under $+, \times$, and $-$ but not closed under $\div$.
3. The set of rational numbers is closed under all the arithmetic operations $+, -, \div$, and $\times$.
   In the case of closure under $\div$, division by 0 must be avoided, as the result is either indeterminate or infinity.

Theorem ($\mathbb{Q}$ is dense in $\mathbb{R}$)

For any two real numbers x and y, if $x < y$ then there exists a rational number q such that $x < q < y$.



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Trichotomy

For any x and y, then exactly one of these applies

$$x < y, \qquad x = y, \qquad x > y \tag{1}$$



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Relations on $\mathbb{R}$

1. A binary relation on a set S describes how two elements s_1 and s_2 are related.
2. We use $s_1 \mathcal{R} s_2$ to mean 's_1 *is related to* s_2'.
3. For instance $s_1 \mathcal{R} s_2$ could mean 's_1 *is a friend of* s_2' or 's_1 is less than s_2'.



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4. $\mathcal{R}$ is **transitive** if for any $s_1, s_2, s_3 \in S, s_1 \mathcal{R} s_2$ and $s_2 \mathcal{R} s_3 \implies s_1 \mathcal{R} s_3$.

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A relation which is reflexive, anti-symmetric and transitive is an **order relation**.
For example $(\mathbb{R}, \leq)$ is ordered.



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Definition (Equivalence relation)

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Order relation on $\mathbb{N}$

We define the **order relation** $<$ on $\mathbb{N}$ by

$$m < n, \quad \text{iff there is some} \quad k \in \mathbb{N} \quad \text{such that} \quad m + k = n \, .$$



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Order relation on $\mathbb{Q}$

The order relation on $\mathbb{Q}$ is defined by $a < b$ iff $a - b < 0$.



Inequalities

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3. If $a, b \in \mathbb{R}$ such that $ab > 0$, then either $a > 0$ and $b > 0$ (that is both are positive) or $a < 0$ and $b < 0$ (that is both are negative).

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7. If $a < b$ and $c < 0$ then $ac > bc$
8. If $0 < a < b$ then $\frac{1}{a} > \frac{1}{b}$.

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4. Relations on $\mathbb{R}$
   - Inequalities
5. Absolute value



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The **absolute value** of a real number x is the non-negative real number denoted by $|x|$ such that

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4. $|x| = 0 \Leftrightarrow x = 0$

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   Therefore $|x-2| = \begin{cases} x-2 & \text{if} \quad x \geq 2 \\ -x+2 & \text{if} \quad x < 2. \end{cases}$

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   Since $3 > 1$ then it is a valid solution as well.
7. In summary, the solutions are $\{-1, 3\}$.



Exercise

1. We denote by $\mathbb{Z}_-$ the set of non-positive integers. Fill the table below with True or False

	$(\mathbb{N}, +)$	$(\mathbb{Z}_-, -)$	$(\mathbb{Z}, \times)$	$(\mathbb{Q} \setminus \{0\}, \times)$
Closed				
Commutativity				
Associativity				
Identity exists				
Inverse exists				

2. Let $\propto$ be a binary relation on $\mathbb{R}$ defined by $x \propto y$ if and only if there exists $q \in \mathbb{Q} \setminus \{0\}$ such that $x = qy$. Find the properties of $\propto$. Is this an equivalence or an order relation?

3. Rewrite the following without the absolute value:
   1. $|x + 5| = 10$
   2. $|-x + a| = 3a - 2,$
   3. $-3|1 - 2x| = -9$



END OF LECTURE

THANK YOU

